

**Sana'a
UNIVERSITY**



**FACULTY OF
ENGINEERING**

Mechatronics Department

Second Year

Report

Thermal Simulation in Solid work

Supervision

Prof. Hamod Alnhary

Eng. Asim

Adopted by

Ibrahim Alkhlydy 202073007

Abdulrhman Al-aghbari 202073119

Mechatronics (Parallel)

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➤ **Introduction:**

The experience to learn how we can make simulation for the (conduction or convection or radiation) and make all of this in the Solid Work And from the Solid work I make my project. My project is from cop . The cop made of (glass) and how transmission the heat in the materials

➤ **Objectives:**

The objective of the project how to make simulation in solid work and how the hat transmission in the materials.

➤ **Methodology:**

The methodology is a simulation of the heat transmission in the materials And the calibration of the heat

➤ **Equipment:**

The matter is glass

Procedures:

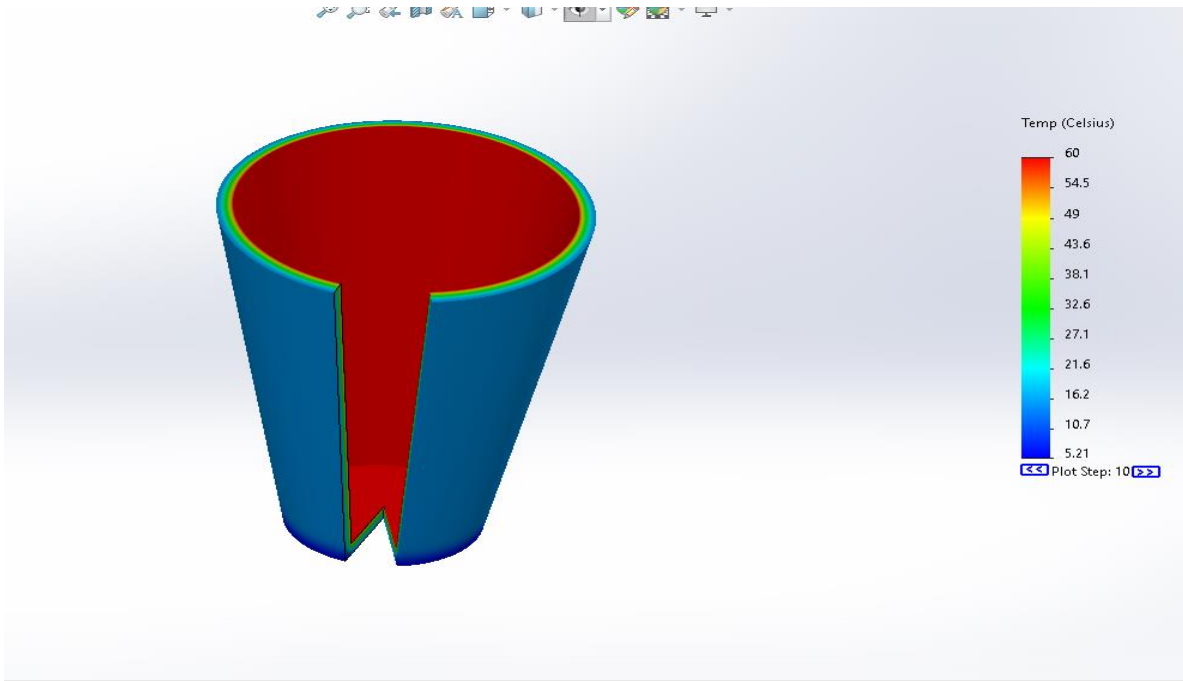
First: Draw the 2D

Second: Draw the 3D

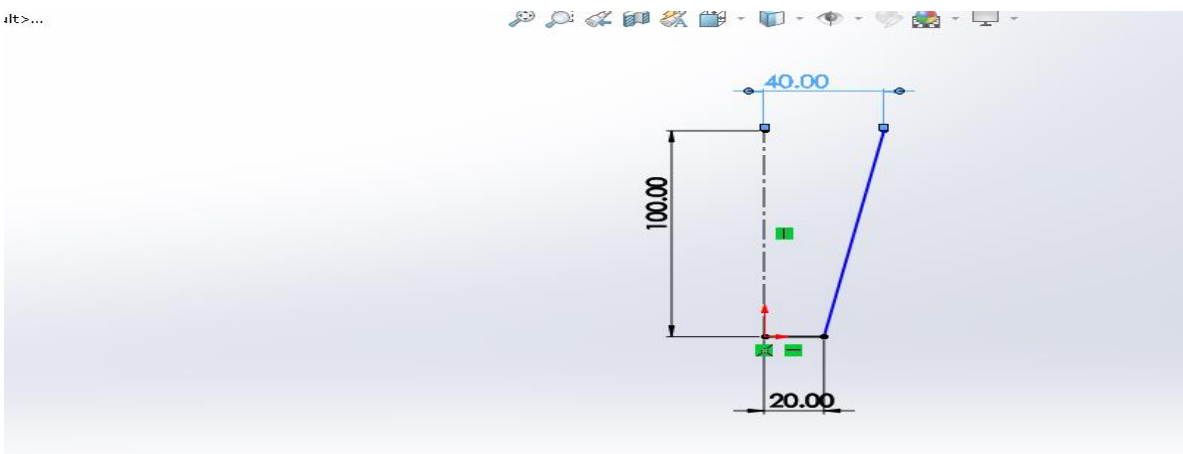
Third: make the thermal conduction

➤ Simulation

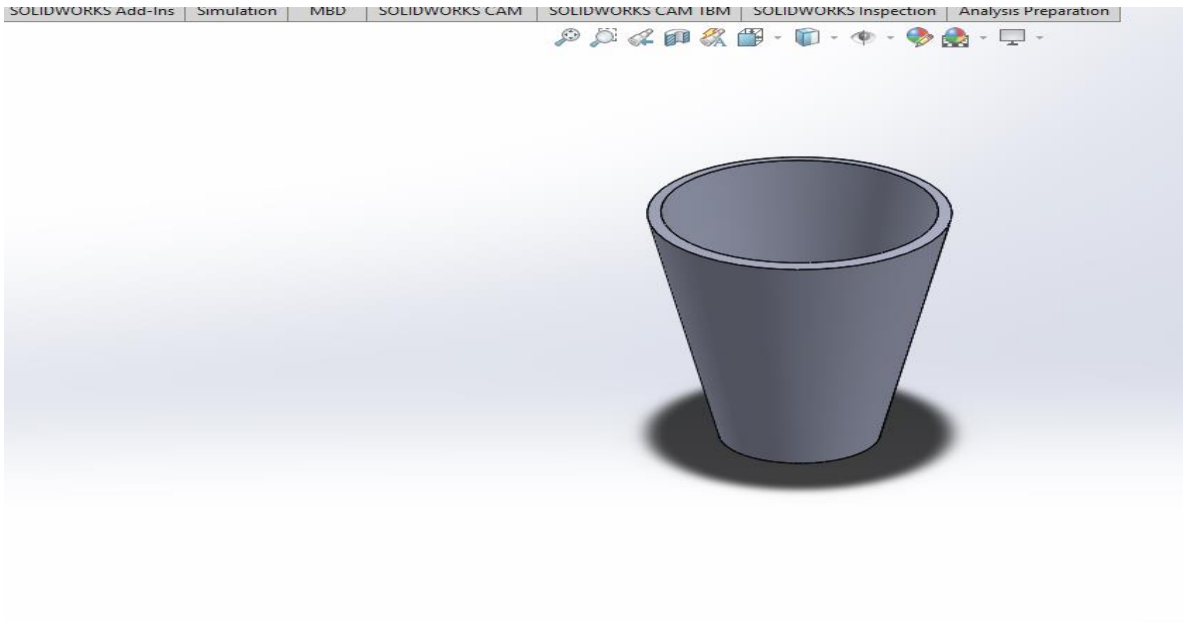
The full project in the solid work



Step one

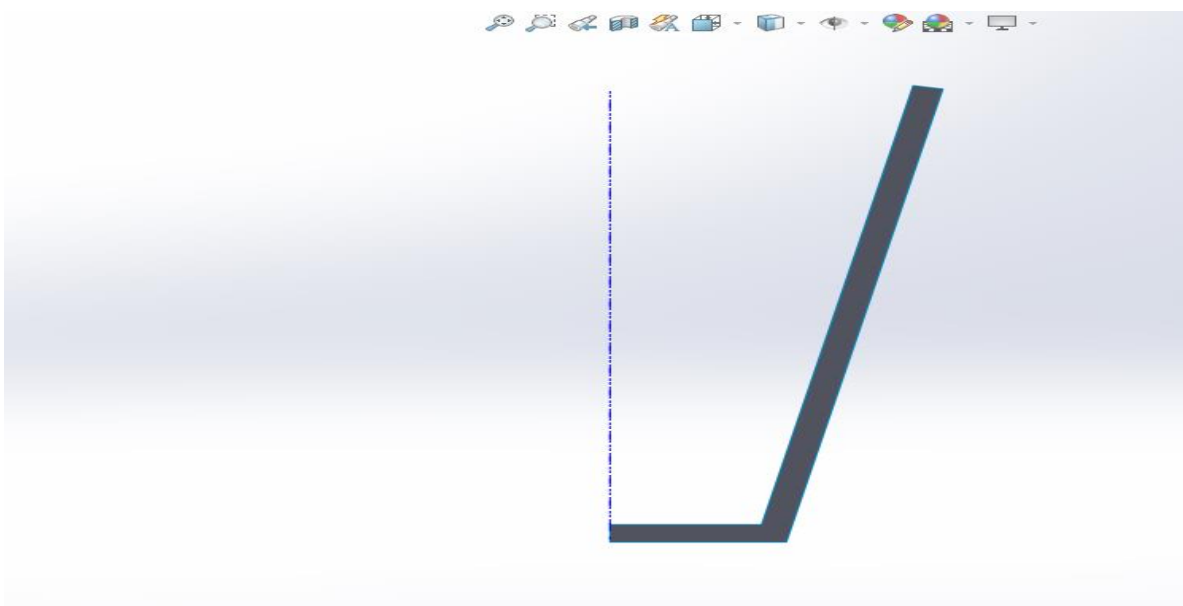


Step two
3D sketch

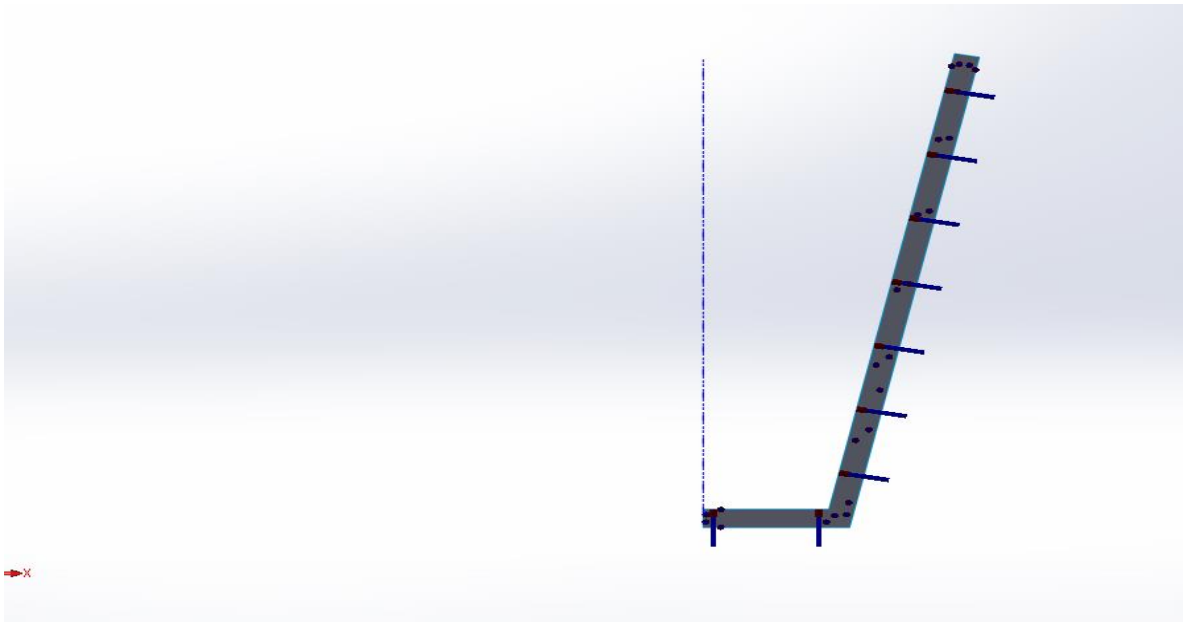


Step three

Make a 2d thermal

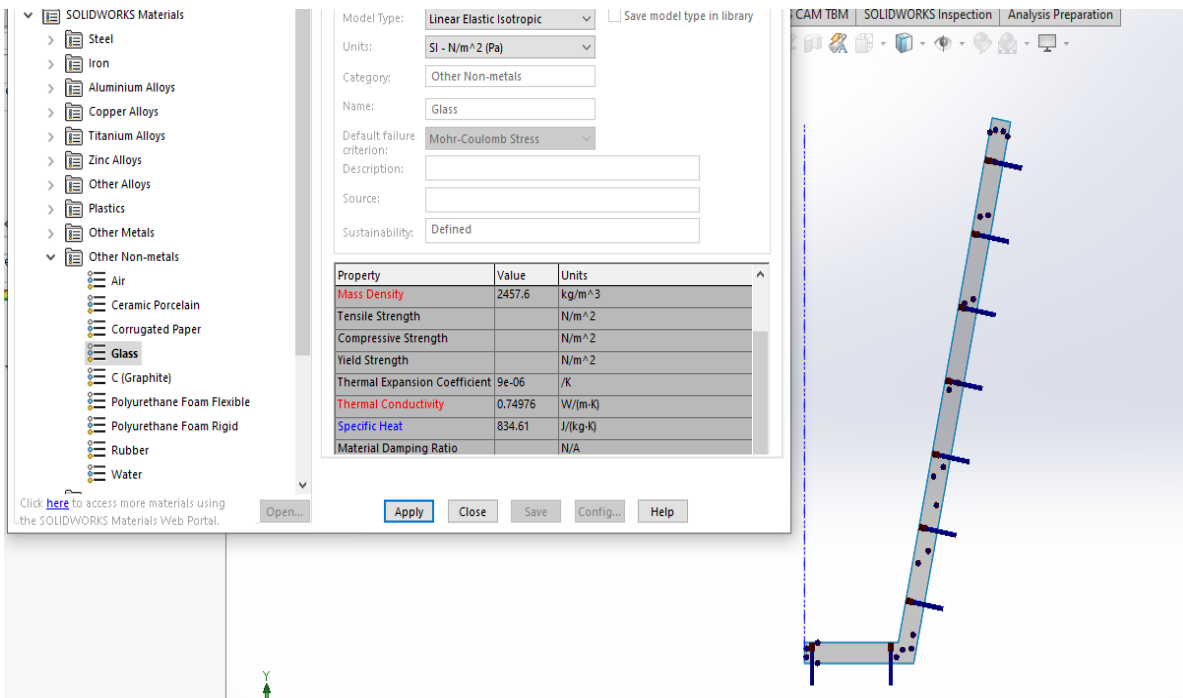


Step four



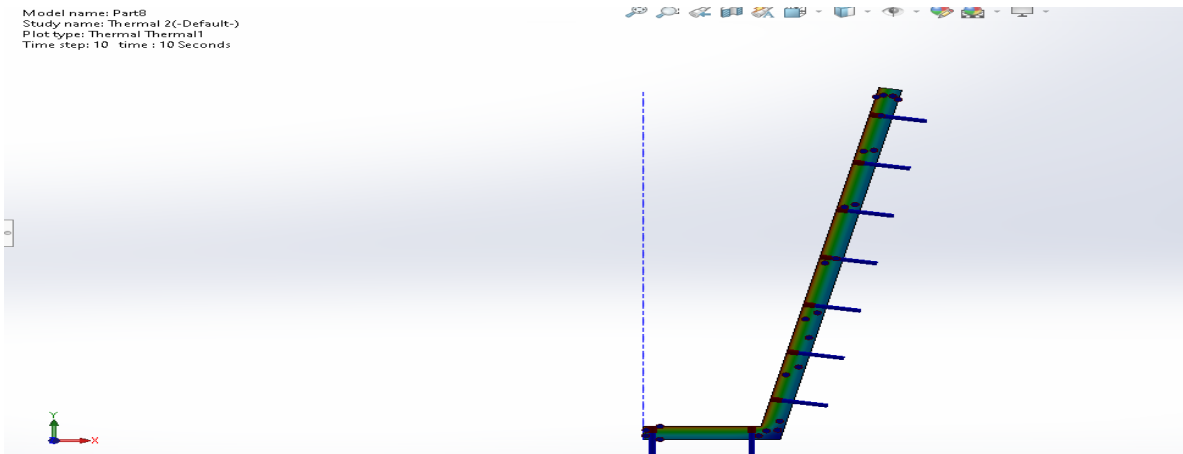
Step five

Property of material



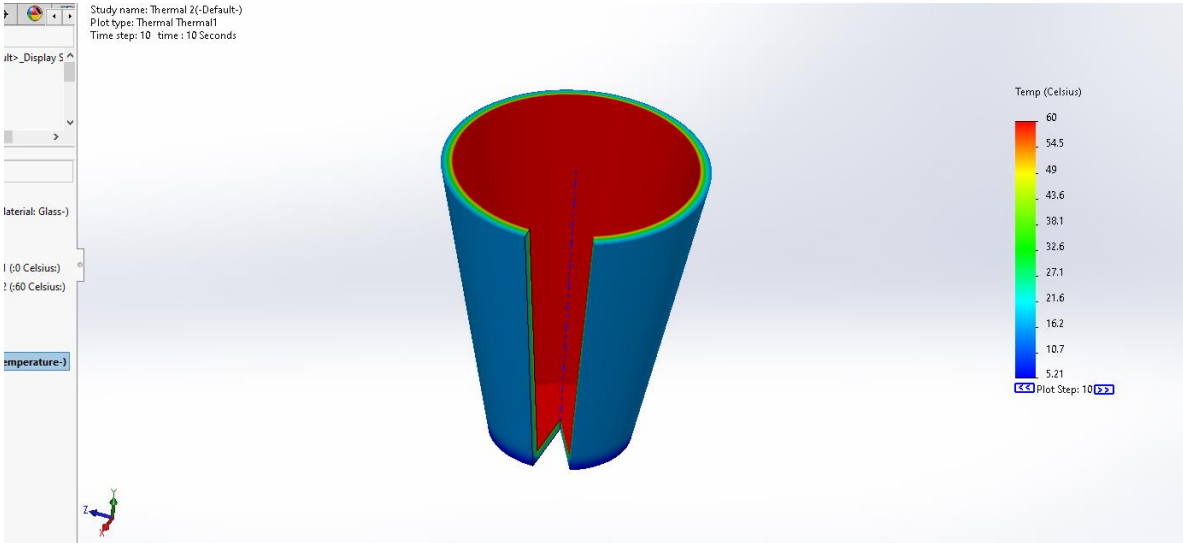
Step Six

Model name: Part8
Study name: Thermal 2(-Default-)
Plot type: Thermal Thermal1
Time step: 10 time : 10 Seconds



Step seven

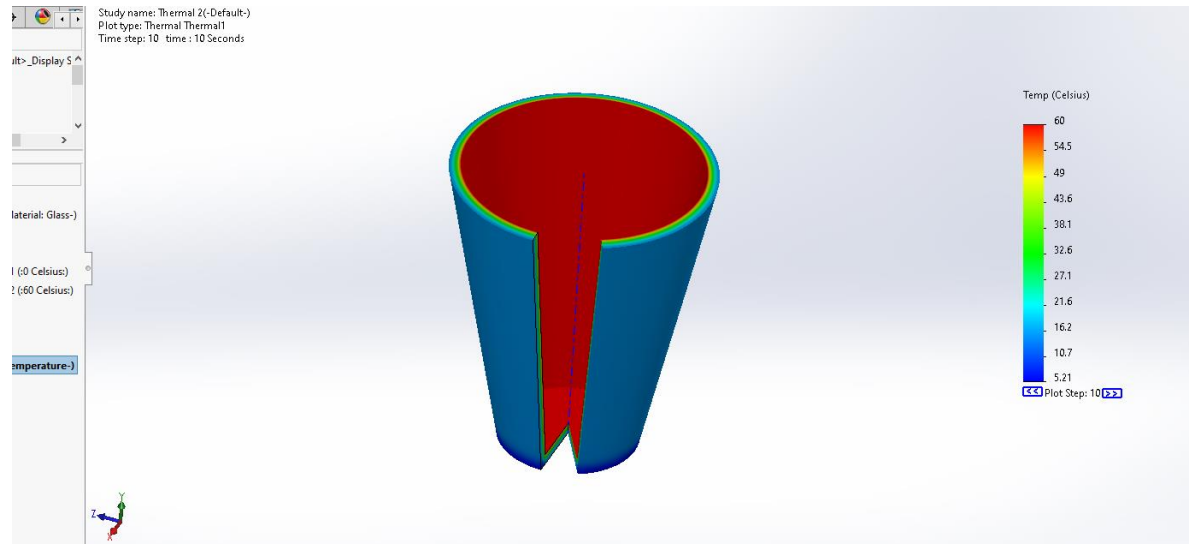
Study name: Thermal 2(-Default-)
Plot type: Thermal Thermal1
Time step: 10 time : 10 Seconds



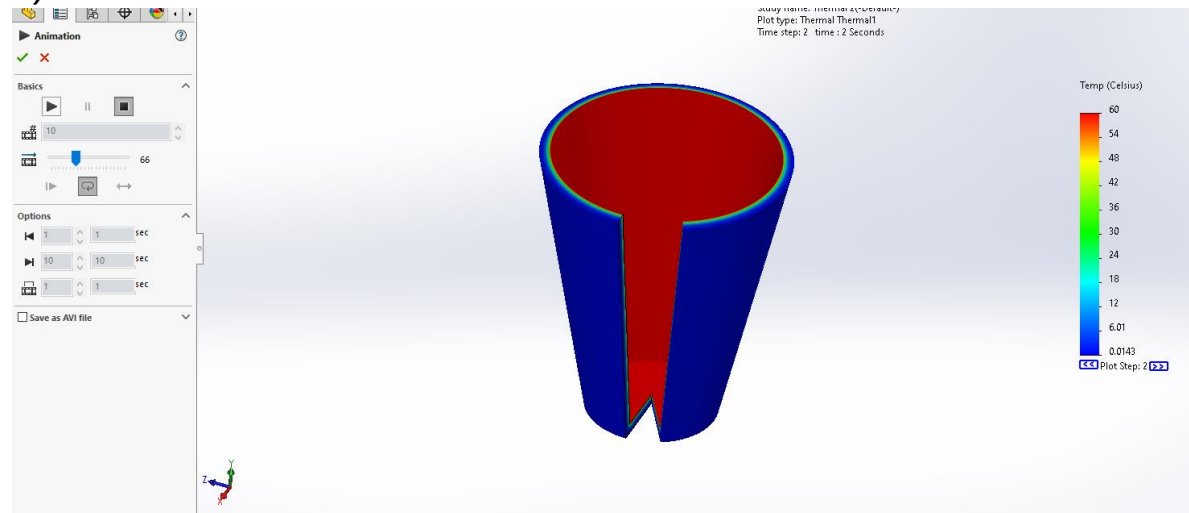
Step eight

Simulation of heat transfer

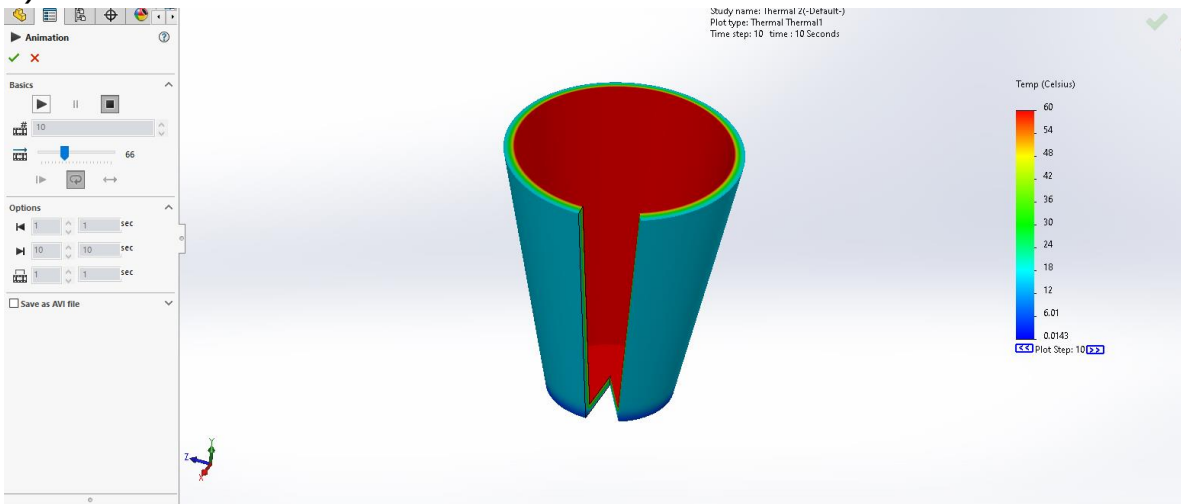
1)



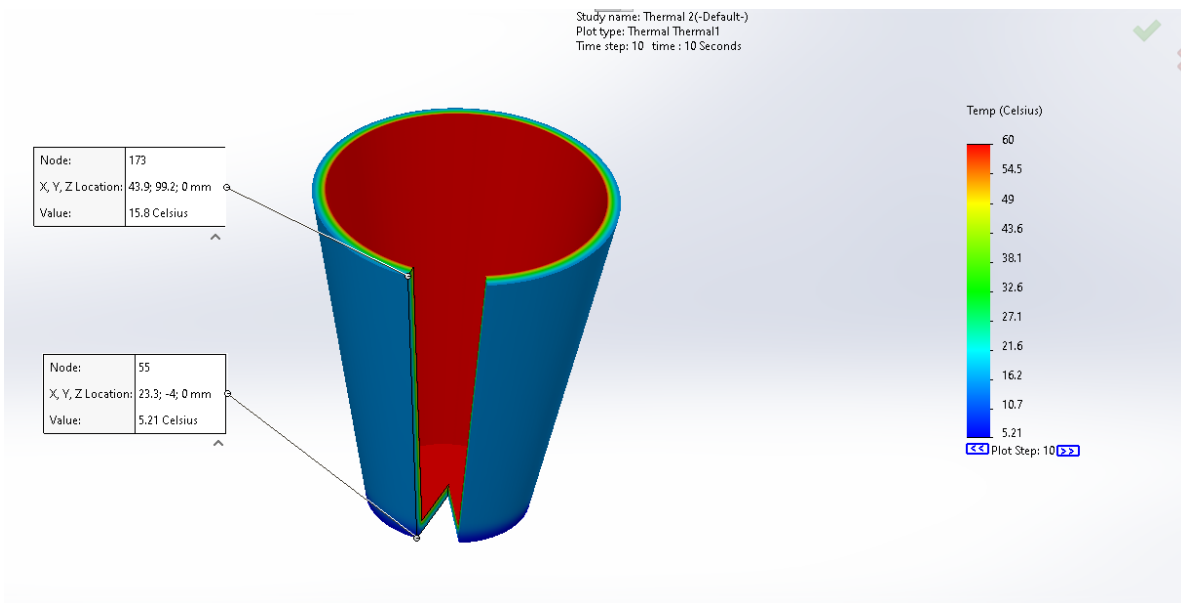
2)



3)

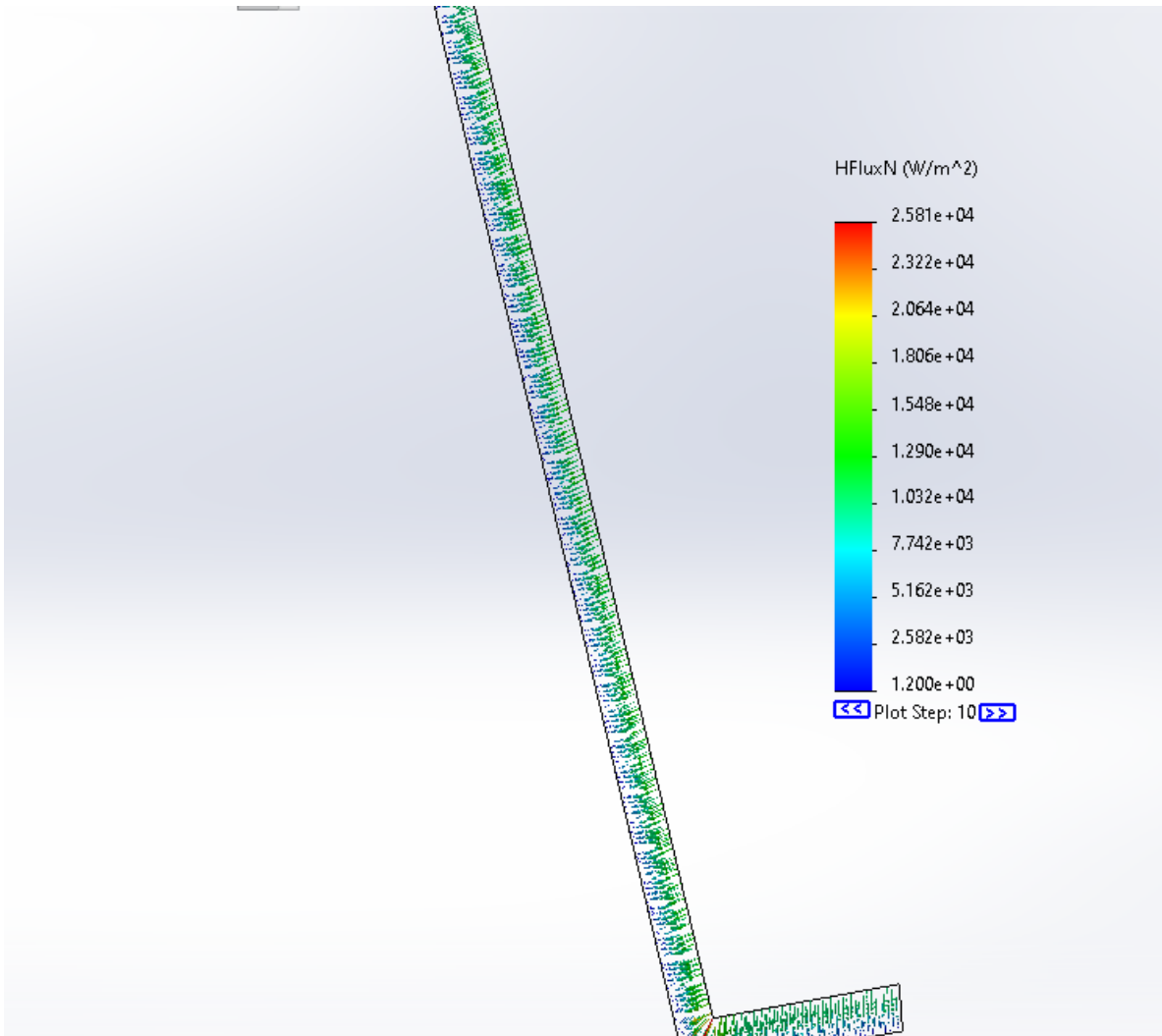


Step nine



Step ten

Simulation of heat



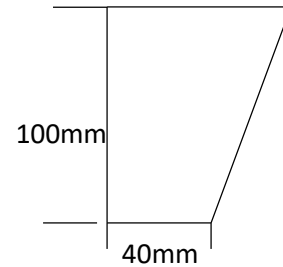
➤ Rules

Q) The (cap) shows a conical section fabricated from (glass). It is of conical cross section having diameter $D = ax^{1/2}$, where $a = 0.7 \text{ m}^{1/2}$. The small end is located at $x_1 = 15 \text{ mm}$ and the large end at $x_2 = 100 \text{ mm}$. The end temperatures are $T_1 = 340 \text{ K}$ and $T_2 = 283 \text{ K}$, while the lateral surface is well insulated. Calculate the heat rate q_x .

KNOWN: Conduction in a conical section with prescribed diameter, D , as a function of x in the form $D = ax^{1/2}$

FIND: (a) Heat transfer rate, q_x

SCHEMATIC:



ASSUMPTIONS: (1) Steady-state conditions, (2) One-dimensional conduction in x -direction

(3) the $T_1 = 340 \text{ K}$, $T_2 = 283 \text{ K}$ (4) $K = 0.8 \text{ W/m}\cdot\text{K}$ (5) $x = 100 \text{ mm}$, $x_2 = 15 \text{ mm}$

PROPERTIES: Glass: $k = 0.8 \text{ W/m}\cdot\text{K}$.

ANALYSIS:

$$Q_x = -kA \frac{dT}{dx} = -k[\pi(ax^{1/2})^2/4] \frac{dT}{dx}$$

Using $A = \pi D^2/4$ where $D = ax^{1/2}$

$$4q_x/\pi a^2 k \sum_{x=1}^x dx/x = \sum_{T=1}^T dt$$

$$q_x = -\frac{\pi}{4} a^2 k (T_1 - T_2) / \ln(x_1/x_2)$$

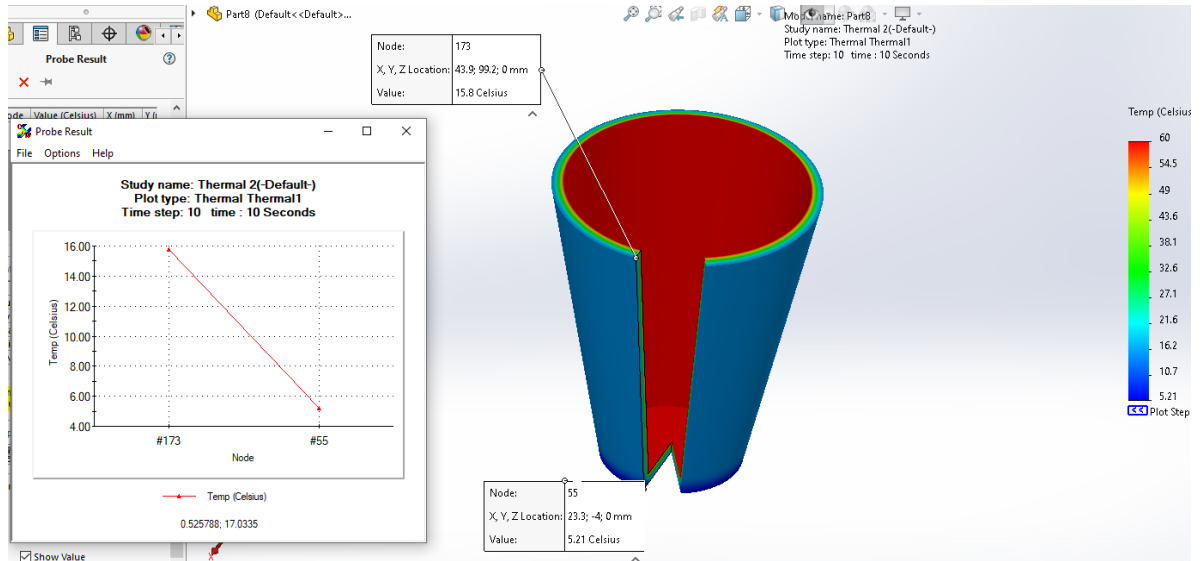
$$T(x) = T_1 + (T_1 - T_2) \frac{\ln(x/x_1)}{\ln(x_1/x_2)}$$

$$T(x) = T_1 - \frac{4q_x}{\pi a^2 k} \ln \frac{x}{x_1} \quad T_2 = T_1 - \frac{4q_x}{\pi a^2 k} \ln \frac{x_2}{x_1}$$

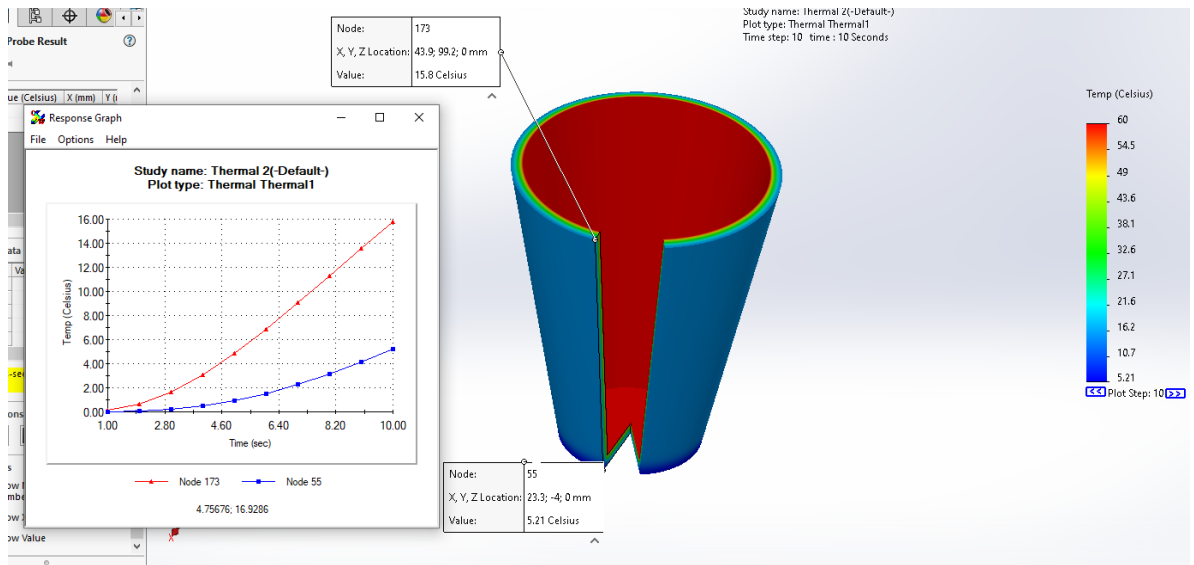
So: $q_x = \pi/4 * 0.7^2 \text{ m} * 0.8(340-283) \ln 15/100 = 33.2 \text{ kW}$

BY hand	BY simulation
$T_1 = 283 \text{ K}$	$T_1 = 278.15 \text{ K}$
$T_2 = 340 \text{ K}$	$T_2 = 333.15 \text{ K}$
$K = 0.8 \text{ W/m}\cdot\text{K}$	$K = 0.74976$

a) Glass



The graph of in and out



Obstacles:

- 1) The first obstacle is when I assembly my project it is nor running I do not know what happened
- 2) The second thing is I do not know how to make a thermal Simulation but in the end I get it
- 3) In the end. The biggest thing is how to choose the mettles
And how to simulation it in solid work

➤ References:

From You top and Google